Yue Lin

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Education

- **The Chinese University of Hong Kong, Shenzhen** — Shenzhen, China
  PhD Student - School of Data Science — 2024.8 - Present
  Research Interests: Multi-Agent Reinforcement Learning, Information Design.
  Fortunately advised by Prof. Baoxiang Wang and Prof. Hongyuan Zha.
  Other close collaborators include Prof. Wenhao Li and Prof. Pascal Poupart.
- **Tianjin Polytechnic University** — Tianjin, China
  Bachelor of Engineering - Computer Science and Technology — 2018.9 - 2022.6
  - **School of Computer Science and Technology** — 2019.9 - 2022.6
    GPA: 3.89/4 (92.2/100) Rank: 1/127
  - **School of Mechanical Engineering** — 2018.9 - 2019.6
    GPA: 3.90/4 (92.0/100) Rank: 1/60

Experience

- **Microsoft** — China
  Research Intern @ Microsoft Research Asia (MSRA) — 2026.8 - Present
  Mentor: Lu Wang [Google Scholar].
- **Tencent** — Shenzhen, China
  Research Intern - Lightspeed Studios — 2025.12 - 2026.6
  Mentor: Yuan Liu.
- **The Chinese University of Hong Kong, Shenzhen** — Shenzhen, China
  Research Assistant - School of Data Science — 2022.2 - 2024.8
  Fortunately advised by Prof. Baoxiang Wang and Prof. Hongyuan Zha.

Publications & Preprints

- Information Design in Multi-Agent Reinforcement Learning.
  Yue Lin, Wenhao Li, Hongyuan Zha, Baoxiang Wang.
  Neural Information Processing Systems (NeurIPS) 2023.
  - Poster. This is one of my representative works.
  - As of 2026, my advisor also regards this as one of his representative works.
  - Sources: [Paper], [Code], [Blog], [Talk].
  - TLDR: Introduces a Markov signaling game and an obedience-constrained signaling-gradient algorithm that learns what information to reveal over time so self-interested receivers are willing to follow the sender's recommendations.
- Verbalized Bayesian Persuasion.
  Wenhao Li, **Yue Lin**, Yun Hua, Xiangfeng Wang, Bo Jin, Hongyuan Zha, Baoxiang Wang.
  International Conference on Machine Learning (ICML) 2026.
  - Sources: [Paper], [arXiv].
  - TLDR: Introduces a verbalized Bayesian-persuasion framework in which LLMs play sender and receiver while an equilibrium solver searches for natural-language messages the receiver has reason to follow, including in multi-stage dialogues.
- Policy-Conditioned Policies for Multi-Agent Task Solving.
  Yue Lin, Shuhui Zhu, Wenhao Li, Ang Li, Dan Qiao, Pascal Poupart, Hongyuan Zha, Baoxiang Wang.
  arXiv preprint. 2025-12-24.
  - This work was scooped by *Evaluating LLMs in Open-Source Games*. We then planned a substantially improved version, but that follow-up direction was also scooped by Google DeepMind's *Code-Space Response Oracles: Generating Interpretable Multi-Agent Policies with Large Language Models*.
  - Sources: [Manuscript].

    - TLDR: Represents each agent's strategy as readable source code and introduces Programmatic Iterated Best Response, where an LLM rewrites one policy against another using game rewards and unit tests as feedback.

- History-Dependent Policy Gradient.
  Yue Lin, Jiacheng Nie, Hongyuan Zha, Baoxiang Wang.
  Draft. 2024.
    - No paper was produced; our method was scooped by Google's paper *Multi-agent cooperation through learning-aware policy gradients* (*ICLR 2025*).
    - TLDR: Derives an exact policy gradient for non-stationary multi-agent learning systems by explicitly accounting for the histories of the other agents.
- The Reciprocity Gradient.
  Yue Lin, Pascal Poupart, Shuhui Zhu, Dan Qiao, Wenhao Li, Yuan Liu, Hongyuan Zha, Baoxiang Wang.
  arXiv preprint. 2026-05-08.
    - This is one of my representative works.
    - Sources: [Manuscript].
    - TLDR: Introduces the reciprocity gradient, which analytically traces how actions and evaluative signals propagate through reputation chains and return as future rewards, jointly optimizing both without reward shaping.
- Talk, Judge, Cooperate: Gossip-Driven Indirect Reciprocity in Self-Interested LLM Agents.
  Shuhui Zhu, **Yue Lin**, Shriya Kaistha, Wenhao Li, Baoxiang Wang, Hongyuan Zha, Gillian K. Hadfield, Pascal Poupart.
  International Conference on Machine Learning (ICML) 2026.
    - Sources: [Paper], [arXiv], [Code].
    - TLDR: Introduces ALIGN, a decentralized mechanism in which self-interested language-model agents exchange open-ended, tone-graded gossip to build reputations, coordinate social norms, and isolate defectors.
- Information Bargaining: Bilateral Commitment in Bayesian Persuasion.
  Yue Lin, Shuhui Zhu, William A Cunningham, Wenhao Li, Pascal Poupart, Hongyuan Zha, Baoxiang Wang.
  EC 2025 Workshop: Information Economics and Large Language Models.
    - Sources: [Paper], [Code].
    - TLDR: Recasts long-term Bayesian persuasion as two-sided bargaining with commitment on both sides, separating the value of private information from the advantage of proposing first.
- Innovative Design and Simulation of a Transformable Robot with Flexibility and Versatility, RHex-T3.
  Yue Lin, Yujia Tian, Yongjiang Xue, Shujun Han, Huaiyu Zhang, Wenxin Lai, Xuan Xiao.
  International Conference on Robotics and Automation (ICRA) 2021.
    - Oral. Delivered a presentation at the Xi'an conference venue.
    - Sources: [Paper], [Blog].
    - TLDR: Introduces a two-degree-of-freedom transformable RHex design that switches among wheel, leg, claw, and hook modes for efficient travel, obstacle crossing, object transport, and ladder climbing.
- A snake-inspired path planning algorithm based on reinforcement learning and self-motion for hyper-redundant manipulators.
  Yue Lin, Jianming Wang, Xuan Xiao, Ji Qu, Fatao Qin.
  International Journal of Advanced Robotic Systems (IJARS) 2022.
    - Sources: [Paper], [Code], [Blog].
    - TLDR: Introduces Swinging Search and Crawling Control, which uses reinforcement learning over the arm's self-motion to find a collision-free target configuration, then moves the arm toward it by crawling instead of searching every intermediate pose.

Patents

- Method and Apparatus for Pretraining Large Language Models Based on Federated Learning (基于联邦学习的大语言模型预训练方法和装置).
  Ang Li, Baoxiang Wang, Hongyuan Zha, **Yue Lin**.
  Chinese Invention Patent.
  No. ZL 2026 1 0720098.0 (Grant No. CN 122242656 B), Granted Jul. 2026.
- Communication Method, Terminal Device, and Storage Medium for Multi-Agent Reinforcement Learning (多智能体强化学习通信方法、终端设备及存储介质).
  Yue Lin, Wenhao Li, Hongyuan Zha, Baoxiang Wang.
  Chinese Invention Patent.
  No. ZL 2023 1 0397744.0 (Grant No. CN 116455754 B), Granted Sep. 2025.

Professional Services

- NeurIPS 2025 Top Reviewer
- ICML 2026 Silver Reviewer [Award Email Screenshot]
- Independent reviewer for top-tier venues:
  - NeurIPS 2024 [6; 45615]; 2025 [5; 32912]; 2026 [4; 17738]
  - ICLR 2025 [3; 21831]; 2026 [1; 5994]
  - ICML 2025 [6; 32893]; 2026 [6; 43185]
  - AAAI 2026 [0]
  - TMLR 2025 [2; 38363]; 2026 [2; 10331]
- Volunteer:
  - ACL 2026 [1; 4155]
  - AAMAS 2024 [3, 9876], 2025 [2, 4792], 2026 [2, 2303]
  - ICML (Position) 2025 [2, 5016]
- Numbers in brackets indicate the number of manuscripts reviewed and the character count of all reviews, respectively. Total reviews: 45.
- Served as a teaching assistant for a postgraduate course on Artificial Intelligence.